

PRAGMATIC REWARDS: INTRINSIC COMMUNICATIVE CLARITY IN MULTI-AGENT REINFORCEMENT LEARNING

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ABSTRACT

Reinforcement Learning (RL) agents often fail in cooperative settings because their actions, while individually optimal, are unpredictable to their partners. This challenge is magnified in human-agent teams, where an agent’s opacity can erode trust and team effectiveness. We address this limitation by establishing a formal equivalence between pragmatic inference, as formalized by the Rational Speech Act (RSA) framework, and Maximum Entropy Reinforcement Learning (Max-Ent RL). This theoretical bridge yields a principled intrinsic reward for “legible” behavior—actions that clearly communicate an agent’s private state—computed offline by a Pragmatic Reward Module (PRM) that adds no latency to decision-making. Crucially, the framework exposes the literal listener L_0 , the assumed common ground against which clarity is measured, as its central design choice. We study a spectrum of listeners, from hand-crafted geometric heuristics to a listener learned jointly with the policy, and show that the learned instantiation turns the pragmatic reward into a variational lower bound on the mutual information between an agent’s intention and its observable behavior, unifying pragmatic and information-theoretic approaches to implicit communication. Experiments in a cooperative navigation task with private preferences show that hand-crafted listeners induce costly signaling that fails to translate into coordination, whereas the learned listener produces calibrated, legible behavior. Our framework provides a formal model for agents that are not only effective but also strategically collaborative.

Index Terms— Reinforcement Learning, Intrinsic Motivation, Multi-Agent Systems, Computational Pragmatics, Legibility

1. INTRODUCTION

Human collaboration is fundamentally a goal-oriented, strategic activity. To succeed, we do not just act to achieve our own goals; we act in ways that allow our partners to infer our beliefs and intentions, enabling better coordination. The same is true for artificial agents. In

cooperative multi-agent reinforcement learning (MARL) environments, an agent’s success depends not only on its own actions but also on being understood by its teammates. Consider a cooperative navigation task in which each agent must cover a target of a preferred type: an agent that simply rushes toward its target may take a path that is ambiguous from a teammate’s perspective, causing the teammate to abandon a target the agent never intended to claim. A more sophisticated agent might take a slightly suboptimal path that unambiguously singles out its intended target, making its intention clear and enabling better team coordination.

This property of being easily understood is called “legibility” [1]. However, standard RL agents, optimized solely for an extrinsic task reward, have no incentive to be legible. This leads to our central question: how can we formally encourage an RL agent to choose actions that are not just extrinsically optimal for the task, but also intrinsically optimal for communicating its private state or intentions?

This work proposes a solution in the form of an intrinsic reward, R_{comm} , that quantifies the communicative clarity of an agent’s actions. To calculate this reward, we model an “observer” (e.g., another agent) who uses pragmatic inference to reason about why the agent chose a particular action over the available alternatives. We find the ideal formalism for this observer in the Rational Speech Act (RSA) framework [2], which provides a formal account of reasoning about communicative intent.

Any such observer model must be anchored in a literal listener L_0 : the assumed common ground that specifies how behavior would be interpreted in the absence of pragmatic reasoning. Our analysis and experiments show that this choice is not an implementation detail but the decisive factor in whether pragmatic rewards help or hurt. Hand-crafted listeners can be systematically misaligned with what the policy can express, incentivizing costly signaling that an actual observer cannot decode. This motivates our second contribution: replacing the hand-crafted L_0 with a listener learned from the agents’ own behavior, which guarantees, by construction, that rewarded actions are decodable.

Our contributions are as follows:

- We establish a formal equivalence between the objective functions of Max-Ent RL and the RSA framework, allowing the RSA “listener value” to be imported as a principled intrinsic reward for legibility, and we show that purely communicative models such as CRSA [3] are recovered as a myopic special case ($\gamma = 0$, $R_{\text{ext}} = 0$).
- We operationalize the theory as a Pragmatic Reward Module (PRM): an offline component, compatible with centralized training and decentralized execution (CTDE), that computes R_{comm} from replayed transitions and adds no latency at execution time.
- We characterize the space of literal listeners and prove that instantiating L_0 with a listener trained by maximum likelihood on the agents’ behavior makes the expected pragmatic reward a variational lower bound on the mutual information $I(M; S, A)$ between private intentions and observable behavior. This connects pragmatic rewards to information-regularization [4] and skill-discriminability [5] objectives, and positions them as the literal (0-th order) special case of our framework.
- We evaluate the resulting reward family in a cooperative navigation task with private preferences. Hand-crafted listeners replicate a failure mode we term miscalibrated signaling—agents pay a task-reward cost for signals that carry no decodable information—while the learned listener yields legible behavior at little or no task cost.

2. RELATED WORK

Our framework, which uses pragmatic inference to generate an intrinsic reward for communicative clarity, builds on four lines of research: (1) legibility and observer-aware planning; (2) intrinsic rewards for multi-agent coordination; (3) implicit and emergent communication; and (4) computational pragmatics.

2.1. Legibility, Goal Recognition, and Observer-Aware Planning

Inferring an agent’s latent state (e.g., its goal) from observed behavior is the central problem of Inverse Reinforcement Learning (IRL) [6] and computational Theory of Mind (ToM) [7, 8]. Legibility inverts this problem: the actor shapes its own behavior so that an observer’s inference succeeds. Originally formalized for robot motion [1], legibility has been extended to stochastic sequential decision problems [9], unified with explicability

and predictability in observer-aware MDPs [10, 11], and recently brought to MARL through legible variants of interactive POMDPs [12]. These approaches typically embed the observer model in the planning objective, requiring inference over goals at decision time. In contrast, our framework converts legibility into an intrinsic reward computed offline during training, so the executed policy carries its communicative competence “for free,” with no observer model, belief tracking, or additional computation in the execution loop.

A complementary line of work uses ToM as a source of intrinsic motivation [13], rewarding agents for accurately predicting the beliefs of others; this is listener-centric, rewarding agents for being predictive of others. Our pragmatic framework is speaker-centric: it rewards agents for making their own private state clear, regardless of their partners’ modeling ability.

2.2. Intrinsic Rewards for Multi-Agent Coordination

Our pragmatic reward is a form of intrinsic motivation for cooperation, and is thus related to social influence [14], which rewards an agent for causally changing its partners’ behavior. The communicative goals differ: influence rewards an agent for compelling a reaction, whereas the pragmatic reward incentivizes the successful transmission of information about oneself, decoupling the speaker’s reward from the listener’s reaction. Teammates remain free to exploit or ignore the revealed intention, which promotes a more flexible form of coordination.

Closest to our learned-listener instantiation is the information-regularization framework of Strouse et al. [4], which adds a mutual-information term $I(A; G|S)$ to the objective so that agents share (or hide) their goal G through their actions. Similarly, Eccles et al. [15] add a positive-signaling bias encouraging messages to depend informatively on observations, and, in the single-agent setting, DIAYN [5] rewards skills for being discriminable from states via a learned classifier. Proposition 1 makes the relationship precise: these MI-based objectives correspond to the literal listener level of our framework, obtained when the RSA recursion depth is zero and L_0 is trained by maximum likelihood. The pragmatic levels add contrastive reasoning over the alternative actions the agent could have taken—a model of communication richer than statistical dependence alone. Our framework also shares the goal of grounded, convention-free coordination with Bayesian action decoding [16] and off-belief learning [18], which control how much agents may read into their partners’ actions; we instead directly reward the actor for making the intended reading easy.

2.3. Implicit Communication vs. Explicit Channels

A significant body of MARL research equips agents with a dedicated communication channel and learns an emergent protocol end-to-end [19, 20, 21]. Our framework diverges from this paradigm by focusing on implicit communication: there is no separate channel and no explicit message passing. An agent “speaks” by moving in the environment, and a teammate “listens” by observing that movement. The pragmatic reward incentivizes actions that are simultaneously effective for the task and clear as signals of intent. This is valuable when explicit communication is costly, unavailable, or must be fused with physical action, as is common in human-agent teams. Because the signal is the action, the exploration-communication conflict identified in emergent-communication work [15, 17] appears here as a reward-shaping trade-off, which we analyze through the weight λ .

2.4. Computational Pragmatics

RSA [2] formalizes communication as recursive probabilistic reasoning between a speaker and a listener, and has been given a rate-distortion interpretation [22]. Neural instantiations replace the hand-specified literal listener with learned models [23, 24], a design choice our learned- L_θ instantiation transposes to the behavioral domain. The Collaborative RSA (CRSA) model [3] extends RSA to multi-turn dialog between agents with asymmetric private information. We show in Section 4.6 that such purely communicative models arise as the myopic, task-free special case of our strategic framework, which extends pragmatic reasoning to long-horizon sequential decision-making where instrumental and communicative goals are intertwined.

3. FORMALISMS FOR ACTION AND INTERPRETATION

To model legible strategic behavior we require two formalisms: one for optimal sequential decision-making under uncertainty, and one for how an observer infers intent from an action. For the former we use Max-Ent RL; for the latter, RSA.

3.1. Maximum Entropy RL

Max-Ent RL extends the standard RL objective with the intuition “be as versatile as possible while still being successful.” In a single-step problem, the agent seeks a policy $\pi(a|s)$ maximizing expected reward plus policy entropy [25]:

$$J(\pi) = \mathbb{E}_{a \sim \pi(\cdot|s)}[R(s, a)] + \alpha_{\text{RL}} H(\pi(\cdot|s)). \quad (1)$$

The optimal policy is a softmax over rewards [26], with temperature α_{RL} trading off exploitation against versatility:

$$\pi^*(a|s) \propto \exp(R(s, a)/\alpha_{\text{RL}}). \quad (2)$$

3.2. The Rational Speech Act Framework

RSA [2] models an observer who infers an agent’s intended meaning not from the literal content of an action alone, but by reasoning about why the agent chose that action over the alternatives: “if they had intended X , they would have done Y ; since they did Z , they must intend something else.” Formally, RSA seeks a communicative policy $S(u|m)$ that optimally balances communicative success against action cost, yielding a softmax over the utility, or “listener value,” of each communicative act:

$$S(u|m) \propto \exp(\alpha_{\text{RSA}} \cdot V_L(u, m)), \quad (3)$$

where $V_L(u, m)$ quantifies the clarity of act u for conveying meaning m to an observer, penalized by the act’s cost.

4. REWARDING LEGIBILITY

Our agent optimizes a composite reward balancing task objectives with communicative clarity. This section defines the reward via a formal bridge between pragmatic inference and RL, characterizes the space of listener models that anchor it, describes the architecture that computes it, and presents the resulting strategic Bellman equation.

4.1. The Total Reward

At each timestep t the agent receives the standard extrinsic task reward $R_{\text{ext}}(s_t, a_t)$ from the environment, and generates internally an intrinsic communicative reward $R_{\text{comm}}(s_t, a_t)$ quantifying how clearly action a_t communicates the agent’s private state m_t (its “meaning,” e.g., an intended target) to an imagined observer. The policy maximizes the weighted sum

$$R_{\text{total}}(s_t, a_t) = R_{\text{ext}}(s_t, a_t) + \lambda R_{\text{comm}}(s_t, a_t), \quad (4)$$

where λ balances task success against communicative clarity. The central question is how to define R_{comm} in a principled way.

4.2. Deriving R_{comm} via Pragmatic Inference

We treat the agent’s action a as a communicative “utterance” and its private state m as the “meaning” to be conveyed (Table 1). At first glance, the objective of an RL agent and the goal of a pragmatic speaker seem

Table 1. Mapping between Max-Ent RL and RSA components.

Max-Ent RL component	RSA analogue
State (s)	Context
Action (a)	Utterance (u)
Private state / intention (m)	Meaning (m)
Agent policy ($\pi(a s, m)$)	Speaker ($S(u m)$)
Intrinsic reward (R_{comm})	Listener value (V_L)
Temperature (α_{RL})	Rationality ($1/\alpha_{\text{RSA}}$)

unrelated. However, the optimal single-step Max-Ent policy (Eq. 2) and the RSA speaker (Eq. 3) are both softmax distributions, identical under the re-indexing $\alpha_{\text{RSA}} = 1/\alpha_{\text{RL}}$. This formal alignment is the cornerstone of our framework: it justifies importing the well-defined listener value V_L from computational pragmatics as a principled intrinsic reward. We define

$$R_{\text{comm}}(s_t, a_t) \triangleq V_L(a_t, m_t) = \log L_t^*(m_t|a_t, s_t) - C(a_t), \quad (5)$$

where $L_t^*(m_t|a_t, s_t)$ is the probability a pragmatic listener assigns to the agent’s true private state after observing its action in context s_t , and $C(a_t)$ is an optional action cost. In practice we set $C(a_t) = 0$: the cost of a legible action is already handled by the task reward, since a less direct but clearer path is naturally penalized by a lower R_{ext} . In our implementation we also add the constant $\log |\mathcal{M}|$, so that R_{comm} reads as the information gain over a uniform prior: it is zero when the action is uninformative and positive when the listener’s posterior concentrates on the true meaning.

4.3. The Space of Literal Listeners

Computing L^* requires anchoring the pragmatic recursion in a literal listener $L_0(m|a, s)$: the assumed common ground specifying how behavior would be interpreted absent pragmatic reasoning. Given L_0 and a set $\mathcal{A}$ of alternative actions (the true action plus K sampled alternatives), the pragmatic listener is obtained by N_r steps of RSA alternation,

$$S_k(a|m) = \frac{L_{k-1}(m|a)}{\sum_{a' \in \mathcal{A}} L_{k-1}(m|a')}, \quad L_k(m|a) = \frac{S_k(a|m)}{\sum_{m'} S_k(a|m')}, \quad (6)$$

and $L^* = L_{N_r}$. The choice of L_0 is the framework’s most consequential free parameter, and we study three instantiations.

(i) Hand-crafted geometric listener. A computationally cheap heuristic interprets actions by their geometric similarity to ideal paths. For agent position p and each meaning m , let v_m be the unit vector toward

the nearest target consistent with m . A naïve score is $S(a, m) = \alpha_{\text{temp}} \cos(a, v_m)$. To capture unambiguity explicitly, an exclusivity variant penalizes similarity to competing meanings,

$$S_{\text{excl}}(a, m) = \alpha_{\text{temp}} (\cos(a, v_m) - \max_{m' \neq m} \cos(a, v_{m'})), \quad (7)$$

and $L_0(\cdot|a) = \text{softmax}_m S_{\text{excl}}(a, \cdot)$. This listener rewards actions aligned with the ideal path toward the true target and misaligned with the ideal paths of all confusable targets.

(ii) Learned listener. Hand-crafted listeners presuppose that the features they score are the ones an actual observer would use. To remove this assumption, we train a network $L_\theta(m|s, a)$ by maximum likelihood on the replay buffer to predict an agent’s true meaning from its publicly observable context and action (the private state is masked from the input). The listener is trained concurrently with the policy, so the common ground co-adapts: the reward only credits signals that are actually decodable from the agents’ current behavioral distribution. With $N_r = 0$ the reward is simply $R_{\text{comm}} = \log L_\theta(m_t|s_t, a_t) + \log |\mathcal{M}|$; with $N_r \geq 1$ the RSA recursion (Eq. 6) is applied on top of L_θ , adding contrastive reasoning over alternatives.

The learned instantiation admits an information-theoretic characterization.

Proposition 1. Let meanings be drawn uniformly on $\mathcal{M}$, and let $L_\theta(m|s, a)$ be any conditional distribution. Then, under the joint distribution induced by the policy,

$$\mathbb{E}[\log L_\theta(m|s_t, a_t) + \log |\mathcal{M}|] \leq I(M; S_t, A_t), \quad (8)$$

with equality iff $L_\theta(m|s, a) = p(m|s_t = s, a_t = a)$. Hence the expected learned-listener reward is the Barber–Agakov variational lower bound [27] on the mutual information between the agent’s private meaning and its observable behavior; maximum-likelihood training of θ tightens the bound, and policy optimization against it maximizes information disclosure.

Proof. $\mathbb{E}[\log L_\theta(m|s, a)] = \mathbb{E}[\log p(m|s, a)] - \mathbb{E}_{s,a}[\text{KL}(p(\cdot|s, a) \| L_\theta(\cdot|s, a)) - H(M|S_t, A_t)]$. Adding $H(M) = \log |\mathcal{M}|$ gives the claim; the KL term vanishes iff L_θ matches the true posterior. $\square$

Proposition 1 unifies two families of methods. The cooperative information regularizer of Strouse et al. [4], the positive-signaling bias of Eccles et al. [15], and the skill discriminator of DIAYN [5] all maximize variational MI bounds of exactly this form; within our framework they are the literal ($N_r = 0$) level of pragmatic reasoning with a learned common ground. Conversely, the RSA recursion is a strict generalization that scores an action not by how much information it carries in isolation, but by how it contrasts with the alternatives the agent declined—the signature of pragmatic communication [2, 22].

4.4. The Pragmatic Reward Module

The theoretical bridge yields a practical hybrid architecture that decouples action selection from communicative analysis. The system comprises the main RL agent, which learns $\pi(a_t|s_t)$, and a Pragmatic Reward Module (PRM), which computes R_{comm} to shape the agent’s learning updates. The pragmatic analysis is performed offline during training—for static listeners, when a transition is collected; for the learned listener, when it is replayed, so the reward tracks the co-adapting common ground—and never in the action-selection loop, adding no latency to decision-making. We operate under CTDE: during centralized training the PRM has access to all agents’ private states, enabling a gold-standard legibility reward; at execution, agents act from local observations only, and the PRM is discarded.

4.5. The Strategic Bellman Equation

With the composite reward defined, the action-value function of the legible agent satisfies the soft Bellman equation

$$Q_t(s_t, a_t) = R_{\text{total}}(s_t, a_t) + \gamma \mathbb{E}_{s_{t+1}} \left[\alpha \log \sum_{a_{t+1}} \exp \left(\frac{1}{\alpha} Q_{t+1}(s_{t+1}, a_{t+1}) \right) \right], \quad (9)$$

which is the standard Max-Ent backup applied to R_{total} . The equation makes explicit the trade-off between immediate gains (task and clarity) and the long-term strategic value of future states: an agent may accept a lower immediate reward (e.g., a “corralling” detour) if the resulting state carries higher expected value through improved team coordination.

4.6. Generality: Recovering Communicative Agents

Purely communicative agents arise as a special case. Setting $R_{\text{ext}} = 0$ and $\lambda = 1$ makes the total reward purely communicative; setting $\gamma = 0$ makes the agent myopic, collapsing Eq. 9 to the immediate reward plus policy entropy. This objective is exactly the per-turn gain function of the Collaborative RSA model [3], demonstrating that single-turn pragmatic models are a specific instance of our framework, which extends pragmatic reasoning to long-horizon sequential settings where instrumental and communicative goals interact.

5. EXPERIMENTAL SETUP

5.1. Environment

We evaluate on a cooperative navigation task with private preferences, a modified Simple Spread scenario in

the spirit of the Multi-Agent Particle Environments [28]. Three agents and three landmarks occupy a continuous 2D arena; each landmark carries a color (a random permutation of three categories per episode) and each agent holds a private preference m_i for one color, drawn uniformly at random. The team reward at each step comprises (a) a preference bonus for every landmark covered by an agent whose preference matches the landmark’s color, (b) a distance penalty equal to the sum, over landmarks, of the distance to the nearest agent, and (c) a collision penalty per overlapping pair. Actions are continuous 2D forces; episodes last 50 steps. Each agent observes its own position and velocity, its preference (one-hot), landmark relative positions and colors, and teammates’ relative positions and velocities—but not teammates’ preferences. Coordination therefore hinges on inferring who is claiming what from motion alone: an agent must both act on its own preference and legibly signal it, so that teammates can confidently cover the remaining landmarks.

5.2. Models and Conditions

All agents are trained with multi-agent Soft Actor-Critic (MASAC)—parameter-shared Gaussian actors and a centralized twin critic over joint observations and actions, with automatic entropy tuning—the natural multi-agent extension of the Max-Ent framework our theory is built on. We compare six conditions that differ only in the reward and observation content:

- Baseline: $\lambda = 0$; extrinsic reward only.
- Oracle: $\lambda = 0$, but observations include teammates’ true preferences, providing an upper bound in which no inference is needed.
- PRM (exclusivity L_0): the hand-crafted geometric listener with exclusivity scoring (Eq. 7) and one step of RSA recursion over $K = 64$ sampled alternative actions.
- PRM (simple L_0): identical but without the exclusivity term, i.e., plain cosine scoring.
- PRM (learned): the learned listener L_θ at recursion depth $N_r = 0$; L_θ is a 2-layer MLP trained by cross-entropy on replayed transitions, with the private preference masked from its input.
- PRM (learned + RSA): the learned listener with one step of RSA recursion over 16 sampled alternatives.

All PRM conditions use $\lambda = 0.1$ applied to the per-team sum of agents’ communicative rewards. Each condition is run with 3 random seeds for 480k environment steps

(64 vectorized environments). Full hyperparameters are listed in Table 2.

5.3. Evaluation Metrics

We evaluate periodically on 64 held-out episodes with deterministic policies. Task performance is the mean per-step team reward and its components. Team specialization is the fraction of landmarks claimed uniquely (each agent assigned to its nearest landmark); a score of 1 indicates perfect division of labor. Legibility is measured by a fixed probe listener—the simple geometric listener, used identically across all conditions—reporting the accuracy with which an agent’s true preference is recovered from its instantaneous action (probe intent accuracy). Because the probe is fixed and external to all training conditions, improvements reflect genuinely clearer behavior rather than adaptation to the metric. For learned-listener conditions we additionally report L_θ ’s own held-out accuracy, and for all PRM conditions the achieved communicative reward R_{comm} .

6. RESULTS

7. DISCUSSION

7.1. Limitations and Future Work

Several limitations qualify our findings and chart future work. First, the intrinsic reward densifies the learning signal, which can aid learning independently of legibility; our probe-based metrics mitigate but do not eliminate this confound. Second, meanings here are hand-specified discrete variables; extending R_{comm} to continuous, hierarchical, or self-assigned meaning spaces—where the agent learns what is worth communicating, not only how—remains open. Third, our validation is confined to AI-AI teams in 2D particle environments; whether computational legibility transfers to human-perceived legibility must be established with human observers, for instance by fitting L_θ to human intent judgments, thereby aligning the intrinsic reward with what people actually find legible. Fourth, CTDE grants the PRM privileged access to private states during training; fully decentralized variants would need agents to estimate each other’s meanings. Finally, the framework is symmetric in an intriguing way: inverting the sign of λ turns legibility into obfuscation, suggesting applications to privacy-preserving and adversarial settings where agents should remain opaque to opponents while staying legible to allies [4].

8. CONCLUSION

We presented a framework that unifies pragmatic inference and maximum-entropy reinforcement learning, importing the RSA listener value as a principled intrinsic reward for legible behavior. The framework isolates the literal listener—the assumed common ground—as the decisive design choice, and our analysis shows that learning this listener jointly with the policy both aligns the reward with what observers can actually decode and recovers information-theoretic approaches to implicit communication as a special case. Empirically, hand-crafted listeners induce costly but undecodable signaling, whereas the learned listener produces calibrated legibility. We view this as a step toward agents whose competence includes being understood—a prerequisite for fluent coordination with artificial and human partners alike.

A. HYPERPARAMETERS

Table 2. Hyperparameters (all conditions unless noted).

Parameter	Value
Parallel environments	64
Episode length	50
Total environment steps	480k
Actor / critic architecture	MLP 256×256
Learning rate (actor, critic)	3×10^{-4}
Discount γ	0.95
Target smoothing τ	0.005
Batch size / updates per env step	512 / 4
Entropy target	$-\dim(\mathcal{A})$ (auto-tuned)
λ (pragmatic weight)	0.1
Listener L_θ	MLP 128×128 , lr 10^{-3}
Heuristic listener temp. $\alpha_{\text{temp}} / K / N_r$	5 / 64 / 1
Learned+RSA alternatives / N_r	16 / 1
Seeds per condition	3

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